

STM32F4 Series self-test library safety manual

Introduction

This document applies to the X-CUBE-STL self-test library for the STM32F4 Series (order code X-CUBE-STL-F4).

X-CUBE-STL-F4, a software expansion package for STM32Cube, includes an application-independent software test library released by ST to implement a relevant subset of safety mechanisms required by the safety concepts applicable to microcontrollers of the STM32F4 Series.

Thanks to this software expansion package, the end user can reduce the time and effort required to implement the conditions of use needed to use the STM32F4 MCUs in safety applications up to SIL 2 hardware safety integrity level with a HFT=0 scheme. The complex activity needed for the design and verification of the safety mechanisms has been carried out by ST on behalf of its customers, and is guaranteed for completeness and accuracy.

Table 1. Applicable product

Part number	Order code
X-CUBE-STL	X-CUBE-STL-F4



1 About this document

1.1 Purpose and scope

This manual is to be used when X-CUBE-STL-F4 is re-used in development of systems based on microcontrollers of the STM32F4 Series.

It provides a detailed description (comprising functions, constraints and evidence), to make possible an assessment of the integrity of a specific safety function that depends (fully or partially) upon X-CUBE-STL-F4.

Therefore, this manual is part of the user documentation, and is supposed to be delivered to customers or partners when requested.

This manual defines the attributes of X-CUBE-STL-F4, including constraints that the integrator shall be aware of, and take into consideration during its application software development. In particular, it constitutes the vehicle for informing the integrator of X-CUBE-STL-F4 properties, of the reasons for its design, its behavior and its characteristics.

This manual has to be considered as the safety manual for X-CUBE-STL-F4 software, in compliance with IEC61508-3, 7.4.2.12 clause b), and depending clauses.

This document applies to the X-CUBE-STL-F4 software test library (self-test library), and is intended for all people (either internal or external to ST) who need to integrate it in a STM32F4 MCUs used in a wider system implementing safety functions.

It is expected that readers of this document are familiar with the concepts and knowledge indicated in related [Section 2.2 Minimum level of competence](#).

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1.2 Terms and abbreviations

Table 2. Terms and abbreviations

Term	Definition
API	Application programming interface
BSP	BSP: board support package
CMSIS	Cortex® microcontroller software interface standard
CoU	Conditions of use
HAL	Hardware abstraction layer
IEC	International electrotechnical commission
Integrator	The person, department or organization that integrates the software
PST	Process safety time
SCi	Systematic capability level
SILi	Safety integrity level
SM_i	Safety mechanism code
SSx	MCU safe state
STL_SSx	X-CUBE-STL-F4 safe state
TMi	Test module code



1.3 Reference documents

- [1] UM1840, STM32F4 Series safety manual available on www.st.com
- [2] UM2494, STM32F4 Series self-test library user guide
- [3] ES0573, X-CUBE-STL software and documentation errata

1.4 Reference normative

This document is written in compliance with the IEC61508 international norm for functional safety of electrical, electronic and programmable electronic safety-related systems, – Part 3: Software requirements.

The version used as reference is IEC61508:1-7[©] IEC:2010.

1.5 Informative section

This document is the IEC61508-compliant safety manual for X-CUBE-STL-F4 software library. The safety characteristics claimed for X-CUBE-STL-F4 are exposed in detail in [Section 3.1 Development process](#) and [Section 3.2 Diagnostic coverage](#):

- a software development process compliant to IEC61508-3 SC3 (systematic capability) – [Section 3.1](#)
- a verified achievable diagnostic coverage on the specified safety scope – [Section 3.2](#)

Nevertheless, the X-CUBE-STL-F4 has been designed to be used in the framework of the STM32F4 MCU safety concept, more specifically to implement a subset of required conditions of use to allow the use of the STM32F4 itself as part of a system implementing safety functions. As long as STM32F4 Series safety manual (ref. [1]) is only referenced in general and not to a concrete fact, this reference is only purely informative, but otherwise mandatory.

To further clarify the relevance of X-CUBE-STL-F4 safety characteristic within the overall IEC61508 STM32F4 safety concept, it is worth to recap that:

- X-CUBE-STL-F4 achievable diagnostic coverage (as stated in [Section 3.2 Diagnostic coverage](#)) allows, among other requirements at MCU level, the use of STM32F4 MCUs in systems implementing safety functions with hardware safety integrity up to SIL2 for HFT=0 and up to SIL3 for HFT=1
- X-CUBE-STL-F4 SC3 software systematic capability allows the end user to integrate the X-CUBE-STL-F4 itself with application software having equivalent or lower systematic capability, excluding interferences between the X-CUBE-STL-F4 and the application software.

Eventually, it is reminded that STM32F4 Series safety manual [1] is the main reference for the use of the STM32F4 MCUs in systems implementing safety functions. End customers willing to use the STM32F4 MCUs in safety-related applications must refer to such document.



2 Release

2.1 Identification

The version of application-independent software test library, self-test library, available in the X-CUBE-STL-F4 software expansion package (and associated to this manual) is V2.0.0.

Note: *In this document any information and characteristic provided for X-CUBE-STL-F4 must be intended as related to the application-independent software test library included as object code.*

2.2 Minimum level of competence

In the spirit of the recommendation of IEC61508-3 Annex D, D 2.4 a), it is assumed that the integrator (team or individual) of X-CUBE-STL-F4 is familiar with:

- concept of functional safety standards application to microelectronics devices and software applications (in the specific, IEC61508)
- STM32F4 MCU IEC61508-compliant safety concept as described in the related safety manual (refer to [1])
- general STM32F4 MCU architecture, based on an Arm® Cortex®-M4 core.

2.3 Purpose

X-CUBE-STL-F4 includes an application-independent software test library released by ST to implement a relevant subset of safety mechanisms required by the safety concept of STM32F4 MCUs.

Thanks to this software the end user can save time and efforts required to implement the conditions of use needed to achieve up to SIL 2 hardware safety integrity level (with HFT=0) on its application. The complex activity needed for the design and verification of the safety mechanisms has been carried out by ST on behalf of its customers, and is guaranteed for completeness and accuracy.

X-CUBE-STL-F4 satisfies the CoUs of

- CPU_SM_0
- RAM_SM_0
- FLASH_SM_0

CoUs / safety mechanism implementation is available under the form of specific API calls included in the relocatable object code included in X-CUBE-STL-F4. Refer to [2] for details.

The table below maps the characteristics required for the above mentioned safety mechanisms in the X-CUBE-STL-F4 implementation. Reference is the safety mechanism description in [1].

Note: *To better understand the relationship between X-CUBE-STL-F4 features and STM32F4 microcontroller CoUs, it is recommended to carefully read the contents of Section 1.5 Informative section.*

Table 3. Safety mechanisms implementation in X-CUBE-STL-F4

Characteristic	CPU_SM_0	RAM_SM_0	FLASH_SM_0
X-CUBE-STL-F4 function	CPU Arm core tests	RAM tests	Flash memory tests
Detailed implementation	Implemented		
Error reporting	Implemented		
Fault detection time	Implemented		
Addressed fault model	Implemented		
Dependency on MCU configuration	Not needed	X-CUBE-STL-F4 is configurable	
Initialization	Implemented		
Periodicity	Implemented, end user has to guarantee		
Test for the diagnostic	Implemented		
Multiple faults protection	Implemented in X-CUBE-STL-F4 concept	Partially implemented (CPU_SM_1 is under end user responsibility)	Implemented
Recommendations and known limitations	Implemented		



2.4 Limitations

2.4.1 Potential transient fault effects

Despite designed to detect hardware permanent failures, X-CUBE-STL-F4 can be forced by transient faults (also known as soft errors) to “false” failure detection. For instance, bit flips into general purpose registers or program counter during the test execution can lead to wrong signature computation; many other similar examples could be provided. If the PST budget is wide enough, executing a second time X-CUBE-STL-F4 failed check could be a way to discriminate permanent faults from soft errors. Anyway, even in such a case, the caveats stated here below (Fail operational restrictions) must be seriously considered.

2.4.2 Fail operational usage

X-CUBE-STL-F4 is a safety mechanism designed to guarantee fault detection features. If the end user considers using it in a framework of fail operational implementation, the following information must be considered:

- X-CUBE-STL-F4 is based on fault detection – so no any kind of fail operational feature is implemented. Conversely, given its nature, as soon as a fault is detected, the information is propagated to application software (to save time and to help stay into PST allocated budget for fault reaction).
- Given the nature of CPU functions, often overlapped one with the others, the information of fault detection by a CPU test specific module cannot guarantee that such fault actually belong to the related function under test. To give an example, even if a fault is detected by the CPU test module for multiplication function, there it is no guarantee that other CPU functions are not affected as well. Accordingly, when a fault is detected by any CPU test module the end user must consider the CPU as no more correctly working. The implementation of fail operational schemes in case of CPU failures are therefore forbidden.
- Even in case of fault detection by a X-CUBE-STL-F4 module testing SRAM and flash memory (where the information of failed sector is available from test results), a marginal probability that such failure has been caused by a CPU failure still exists. The successfully execution of a full CPU test is therefore highly recommended in schemes where the memory error is managed to implement a fail operational scheme. Beware of potential dangers of wrong fail operational software routines execution due to SRAM and/or flash memory malfunction.
- The library function *artificial failing* (refer to [2] for details) has been implemented in upper software layers of the library with the purpose to provide a way to get error messages from the library itself without external instrumentation. The function is therefore meant to be used to facilitate the verification activity for the management of library error messages in the end-user application software. Accordingly, it cannot be considered as an exhaustive simulation of actual CPU failures on real devices.

2.5 Compatibility

2.5.1 Version history

This section includes information related to each released version, such as the nature of changes between versions and the possible necessity of updates. Refer to [Table 4](#).


Table 4. Library version history

Version	Changes respect to previous version	Notes
2.0.0	- Removed dependency from errata sheet ([3]) which no longer applies to this version. - Removed interrupt disable constraints on TM7L and TM8. - Introduced the ability to select between CPU test segments TM7L and TM7. - Removed some running time constraints related to the CRC hardware module used by the scheduler. - Library compiled with state-of-the-art version of certified compiler EWARMFS 9.20.3.	New designs are recommended to use release 2.0.0.
1.0.0	N/A	The update of existing systems using release 1.0.0 is not necessary. This version of X-CUBE-STL-F4 is affected by a limitation described in the related errata sheet ([3]), which must be considered as an integral part of X-CUBE-STL-F4 safety documentation. The limitation is the following: CPU: the CPU TM9 function may return a wrong test failure status. The end-user must follow the errata sheet guidance to determine if the final application is exposed to the above limitation effect. If so, it is mandatory to apply the related workaround.

2.5.2

Backward compatibility

The software library version 2.0.0 addressed by the present manual is compatible with all previous versions. In case of update of existing projects, the end-user must follow the specific integration guidelines included in a dedicated section of the user guide [2].



2.5.3

Other compatibilities

X-CUBE-STL-F4 is delivered as a dedicated software expansion package for X-CUBE environment. This inclusion is done only for delivery-related reasons – the main core of X-CUBE-STL-F4 is the software test library available as relocatable object code (refer to [2] for details). X-CUBE environment is therefore not mandatory for the use of X-CUBE-STL-F4 in an end customer system.

X-CUBE-STL-F4 is designed as HAL-, BSP- and CMSIS-agnostic; there are no dependencies from these software packages.

X-CUBE-STL-F4 does not include any pre-existing software as defined by IEC61508-4, 3.2.8

The main core of X-CUBE-STL-F4 is based on a relocatable object code. Therefore, there are no dependencies from the compiler used to build the final application software. The end user is free to select the compiler according to its proprietary tool compliance strategy, with no constraints coming from the use of X-CUBE-STL-F4.

X-CUBE-STL-F4 claimed diagnostic coverage is independent from the used compiler.



3 Safety standards compliance

3.1 Development process

The application-independent software test library available in X-CUBE-STL-F4 as object code (refer to [2] for details) is developed according to an internal software development process that ensures a systematic capability at SC 3 (often described as SIL 3), as per IEC61508-3.

The compliance to IEC61508-3 is certified by an independent third party assessor - refer to [Section 3.4](#) for details.

3.2 Diagnostic coverage

X-CUBE-STL-F4 achieved diagnostic coverage on related safety scope has been verified according to an internal process which is compliant to IEC61508-2 requirements. The process includes also verification activities based on fault injection campaigns on actual device netlists by means of state-of-the-art simulation tool suites. The accuracy of diagnostic coverage claims for X-CUBE-STL-F4 is therefore in line and even beyond what required by IEC61508-2.

Evidences of verification process and results are stored in ST premises. Due to presence of ST confidential information, these contents are not publicly available. They are however available for possible competent bodies audit and inspections. This is in line with the clarification of Note 2 on IEC61508:2, 7.4.9.7 requirement.

The verified X-CUBE-STL-F4 diagnostic coverage is detailed in the following table.

Table 5. Diagnostic coverage

Tested module	Verified diagnostic coverage (DC fault model) on STM32F4 Series MCUs ⁽¹⁾
CPU ⁽²⁾	Medium (90%)
Embedded SRAM	Medium (90%)
Embedded flash memory	High (99%)

1. Definitions compliant with Note 2 on IEC61508-2, section A.2 Hardware integrity.

2. On Arm® Cortex®-M4 processor core physical implementation in the STM32F4 MCUs.

The diagnostic coverage has been verified by using - where technically possible - the failure distribution approach and verified for a DC fault model that includes:

- stuck at 0/1 faults
- stuck-open (open or high impedance) faults
- stuck-on faults
- bridging (short circuits) faults (State coupling faults/idempotent coupling faults in SRAM)
- data retention faults (only for SRAM).

Information on the above listed fault models can be found in the state-of-the-art functional safety standard ISO 26262-11:2018.

The compliance to IEC61508 of diagnostic coverage verification method, including above reported claims, is certified by an independent third party assessor - refer to [Section 3.4](#) for details.

Note: X-CUBE-STL-F4 does not address nor mitigate transient failures (also known as soft errors).

The STM32F4 IEC61508-compliant safety concept includes for CPU, SRAM and flash memory additional safety mechanisms to be implemented by the end user. Refer to the section "Condition of use" in STM32F4 safety manual ([1]) for the full list of recommended measures.

Note: As reported in the note on CPU module of [Table 5](#), the implementation of the CPU self-test (CPU_SM_0) is focused mainly on Cortex®-M4 processor core and so it brings a necessary but not exhaustive contribution to achieve a medium diagnostic coverage on the overall CPU subsystem (as it is defined in MCU FMEDA document). To achieve that diagnostic coverage target on the overall CPU subsystem, the user must apply all the other additionally required safety mechanisms focused on the other parts of the subsystem itself like NVIC, MPU, BusMatrix and debug resources, which are listed in the section "Condition of use" in STM32F4 safety manual ([1]).



3.3 Document compliance

This document has been written in compliance with the requirements of IEC61508-3 Annex D. For the sake of clarity Table 6 reports the mapping of this document content versus the requirements listed in the IEC61508-3 Annex D.

Table 6. Mapping of IEC61508-3 Annex D requirements

IEC61508-3 Annex D requirement	Reference(s)
D.1.1/D.1.2/D.1.3	Requirements satisfied by this overall document
D.2.1	Table 7. Mapping of IEC61508-2 Annex D requirements
D.2.2	Section 2.3 Purpose, Section 5 Instructions for use
D.2.3 a)	Section 4 Configuration
D.2.3 b)	Section 4 Configuration
D.2.3 c)	Section 5.1 Installation
D.2.4 a)	Section 1.2 Terms and abbreviations, Section 2 Release
D.2.4 b)	Section 3 Safety standards compliance
D.2.4 c)	Section 5.1 Installation
D.2.4 d)	Section 2.3 Purpose
D.2.4 e)	Section 2.3 Purpose
D.2.4 f)	Section 2.5.2 Backward compatibility
D.2.4 g)	Section 2.5.3 Other compatibilities
D.2.4 h)	Not applicable because X-CUBE-STL-F4 does not include any pre-existing software (as stated in Section 2.5.2 Backward compatibility)
D.2.4 i)	Section 6 Support and maintenance
D.2.4 j)	Not applicable because the X-CUBE-STL-F4 implements all requirements
D.2.4 k)	Section 5.3 Safe state
D.2.4 l)	Section 5.4 Interface constraints
D.2.4 m)	Section 5.5 Security
D.2.4 n)	Section 4 Configuration
D.3.1-2-3	Section 3 Safety standards compliance

Table 7. Mapping of IEC61508-2 Annex D requirements

IEC61508-2 Annex D requirement	Reference(s)
D.2.1 a)	Section 2.3 Purpose and [2]
D.2.1 b)	Section 2.1 Identification
D.2.1 c)	Section 5.2 Assumptions
D.2.2	Not applicable because X-CUBE-STL-F4 is a full software element
D.2.3 a)	Section 3.1 Development process
D.2.3 b)	Section 5.1 Installation, Section 5.2 Assumptions
D.2.4	Refer to Table 6



3.4

Certification

X-CUBE-STL diagnostic software test libraries developed by STMicroelectronics have been successfully assessed by TÜV Rheinland. The certification body verifies that the development process for X-CUBE-STL is compliant to IEC61508 requirements for a systematic capability SC3, and that the verification method for claimed X-CUBE-STL diagnostic coverage is compliant to IEC61508 requirements. The related certificate number is 968/FSP 1627.00/18.

The list of STM32 series addressed by X-CUBE-STL certified software solution is maintained and updated in the document revision list, integral part of certification document set. The revision list also reports the actual version for certified software and related documentation.

The original certificate and the updated revision list can be downloaded from TÜV Rheinland websites: www.fs-products.com, www.certipedia.com.

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4 Configuration

X-CUBE-STL-F4 has several configuration possibilities. The exhaustive list of parameters is reported in [2]. The following sections only describe the configuration options potentially impacting safety characteristics of X-CUBE-STL-F4.

4.1 CPU test modules configuration

The target diagnostic coverage for STM32F4 CPU failure modes is achieved with the execution (at least once per PST) of this combination of test modules:

- TM1L + TM2 + TM3 + TM4 + TM5 + TM6 + TM7L + TM8 + TM9 + TM10 + TM11.

Alternative versions of TM1L and TM7L are available in the library. They can be interesting for customers willing to add some safety margin to their safety concept, or facing safety standards different from IEC61508 and possibly requiring such kind of additional test because of prescriptive requirements:

- TM1L can be replaced by TM1 (full walking bit test on general-purpose registers r0 to r12).
- TM7L can be replaced by TM7 (dedicated test for both stack pointers, MSP and PSP).

The diagnostic coverage value claimed for CPU in [Section 3.2 Diagnostic coverage](#) is achieved regardless of the selection between TM1 and TM1L, and between TM7 and TM7L.

Warning: *The missing execution of one or of several CPU test modules compromises the achievement of the diagnostic coverage value claimed in [Section 3.2 Diagnostic coverage](#) for the Arm® Cortex®-M4 CPU. ST does not plan to provide intermediate safety metrics related to partial execution of the overall CPU test module suite.*

4.2 Safety-related area definition for flash memory and SRAM

The safety analysis outcomes, as reported in [1], allow the user to restrict the application of RAM_SM_0 and FLASH_SM_0 to the portion of memories actually used to store safety-related data and code. Examples of such non safety-related areas could be unused locations, data collected for log/stats, service routines executed only during maintenance and/or off-line operations. The time penalty associated to the periodical execution of SRAM and flash memory test can be therefore reduced and optimized.



5 Instructions for use

5.1 Installation

X-CUBE-STL-F4 correct integration in the final system (including the integration with the application software) must be verified according to the integration guidelines reported in [2]. This verification is a prerequisite for the claims related to X-CUBE-STL-F4 diagnostic capabilities.

The use of a software development flow with systematic capability compliant to SC3 makes it possible to state that X-CUBE-STL-F4 does not interfere with any end user application software executed on the same MCU. At the same time, the end user must guarantee that the application software does not interfere with X-CUBE-STL-F4 operations, by following the prescriptions in [2].

5.2 Assumptions

The following table reports the CoUs for X-CUBE-STL-F4.

The end user must fulfill all the conditions, as any claim concerning diagnostic capability is based on this assumption.

Note: *To better understand the relationship between X-CUBE-STL-F4 assumptions and STM32F4 MCU safety concept, it is recommended to carefully read the contents of [Section 1.5 Informative section](#).*

Table 8. Conditions of use for X-CUBE-STL-F4

ID	Description
CoU_STL_1	It is assumed that X-CUBE-STL-F4 is combined with an external (or internal) watchdog, responsible to reset the MCU and to achieve the local safe state STL_SS2. The associated watchdog must issue a reset to the CPU if the overall test suite is not completely executed at least once per PST ⁽¹⁾ , as specified in CoU_STL_8 ⁽²⁾ .
CoU_STL_2	The end user must integrate X-CUBE-STL-F4 in the application software according to the integration guidelines reported in [2], section "End user integration tests".
CoU_STL_3	The end user is responsible for the transition of the final application to a valid safe state when X-CUBE-STL-F4 reaches its local safe state STL_SS1 or STL_SS2.
CoU_STL_4	It is assumed that Arm® Cortex®-M4 debug features are not used for the implementation of any safety function.
CoU_STL_5	The end user is responsible for the definition of flash memory safety-related area to be checked by X-CUBE-STL-F4.
CoU_STL_6	The end user is responsible for the definition of SRAM safety-related area to be checked by X-CUBE-STL-F4.
CoU_STL_7	The end user is responsible to guarantee the correct coexistence between X-CUBE-STL-F4 and its application SW, according to the guidelines reported in [2], section "STL user constraints".
CoU_STL_8	The X-CUBE-STL-F4 tests suite, defined as the successful execution of: • all CPU test segments⁽³⁾ • Flash memory test over the Flash memory defined safety-related area • SRAM test over the SRAM safety-related area defined by the end user must be completed at least once per system PST ⁽¹⁾ .
CoU_STL_9	The end user is responsible to guarantee the X-CUBE-STL-F4 files integrity by following security related instruction, as described in Section 5.5 Security .
CoU_STL_10	It is highly recommended to execute the overall X-CUBE-STL-F4 test suite as defined in CoU_STL_8 at each system power-up ⁽⁴⁾ .

1. In case of high-demand and low-demand modes of operation, X-CUBE-STL-F4 timing execution constraints follow requirements in Clause 7.4.5.3 of IEC61508-2.

2. The achieved X-CUBE-STL-F4 diagnostic coverage does not depend upon the watchdog implementation.

3. Details on the list of CPU test segments to be executed are provided in [Section 4.1 CPU test modules configuration](#).

4. This measure is introduced to lower the probability of faults accumulation during the time period when the system hosting the STM32F4 Series MCU(s) is not powered.



5.3 Safe state

X-CUBE-STL-F4 local safe state STL_SSx is defined to be compatible with the STM32F4 MCU local safe state SSx defined in [1].

X-CUBE-STL-F4 local safe state is the one in which either:

- STL_SS1: X-CUBE-STL-F4 detects a fault and reports the detection to the application software by the specified result structure
- STL_SS2: the application software cannot be informed by the presence of a fault and the combined watchdog timeout expires, forcing the issue of a reset to the CPU

According to this definition, X-CUBE-STL-F4 and STM32F4 local safe states can be mapped as follows:

- STL_SS1 → SS1
- STL_SS2 → SS2

In other words, the natural evolution of the two X-CUBE-STL-F4 local safe states is the corresponding safe state at STM32F4 MCU level. The application software has the responsibility to drive this transition, as X-CUBE-STL-F4 contribution to the overall system safety is to reach its local safe state (STL_SS1 or STL_SS2).

Note: To better understand the relationship between STM32F4 safe state and X-CUBE-STL-F4 safe state, it is recommended to carefully read the contents of [Section 1.5 Informative section](#).

5.4 Interface constraints

X-CUBE-STL-F4 is delivered as relocatable object code. Interface access can happen only through published API methods. Refer to [2] for API list, parameters meaning, constraints and other related info.

5.5 Security

The integrity of the X-CUBE-STL-F4 object file is protected against unintentional modifications (e.g. download/copy issues) or end user file system malfunctions by a HASH-256 fingerprint. The expected value for Hash fingerprint is reported in the X-CUBE-STL-F4 release note document.

The end user is recommended to check the integrity of the X-CUBE-STL-F4 object file used in its filesystem, according to the procedure reported in [2].



6 Support and maintenance

The end user may contact its local ST office to receive additional support on how to use X-CUBE-STL-F4, if needed.

The end user must contact its local ST office to immediately report any anomaly and/or incorrect behaviour of X-CUBE-STL-F4.

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Revision history

Table 9. Document revision history

Date	Version	Changes
09-Oct-2018	1	Initial release.
22-Nov-2018	2	Updated: Table 8. Conditions of use for X-CUBE-STL-F4
31-Jan-2019	3	Updated: - Section 3.3 Document compliance - Section 4.1 CPU test modules configuration Added: - Table 6. Mapping of IEC61508-3 Annex D requirements
04-Apr-2019	4	Removed former footnote 3 from Section 3.2
02-Sep-2019	5	Updated: - Functional safety documentation framework - Section 2.1 Identification
15-Mar-2022	6	Updated: - Section 1.3 Reference documents - Section 1.5 Informative section - Section 2.3 Purpose - Section 3.2 Diagnostic coverage - Section 5.2 Assumptions
09-Nov-2022	7	Changed self-test library version to 2.0.0. Replaced Flash memory by flash memory in the whole document. Removed caution note from Section 2.3 Purpose. Added details on <i>artificial failing</i> library function in Section 2.4.2 Fail operational usage. Added Section 2.5.1 Version history. Updated Section 2.5.2 Backward compatibility and Section 3.2 Diagnostic coverage. Updated Section 4.1 CPU test modules configuration. Updated note ⁽³⁾ in Table 8. Conditions of use for X-CUBE-STL-F4.



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